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Enhancing PV Array Performance in Partial Shading Conditions: A Comparative Evaluation for MPPT

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***Abstract*— This paper presents a novel method for maximizing power extraction in a photovoltaic (PV) system when subjected to partial shading conditions. The approach integrates multiple optimization algorithms, namely Perturb & Observe (P&O), Particle Swarm Optimization (PSO), Grey Wolf Optimization (GWO), and Firefly Algorithm (FA), to achieve efficient maximum power point tracking (MPPT) in complex shading scenarios. P&O adjusts the duty cycle, while PSO and FA improve population initialization and search speed. GWO presents premature convergence. MATLAB/Simulink simulations validate the approach, demonstrating improved tracking efficiency and system performance compared to other MPPT techniques and intelligent optimization algorithms. After analysis of a PV system, the results show that P&O is inefficient in finding the global maximum power point (GMPP), especially in variable irradiance scenarios. PSO, GWO, and FA successfully track the GMPP, with FA Optimal speed performance, efficiency, convergence, and ripple achieved. The average efficiencies achieved by PSO, GWO, and FA are 96.925%, 97.72%, and 98.51%, respectively. FA also exhibits faster convergence compared to the other algorithms.**

Keywords*— *Maximum Power Point Tracking, Photovoltaic Systems, Partial Shading Conditions, Firefly Algorithm, Grey Wolf Optimization, Particle Swarm Optimization, and Perturb & Observe.

I. Introduction

The PV equivalent circuit serves as a simplified model to comprehend the electrical behavior of PV cells or modules, including elements like a diode, current source, series, and shunt resistors. A crucial method for boosting PV system efficiency is Maximum Power Point Tracking (MPPT). This technique ensures optimal operation by continuously adjusting conditions to extract the maximum available power from solar panels, thus reducing losses and improving overall performance [1]. However, conventional MPPT methods like perturb and observe struggle to accurately pinpoint the global maximum power point in partial shading scenarios caused by factors like nearby structures or vegetation. To tackle these issues, this study introduces a comparative MPPT approach tailored for enhancing PV system performance under partial shading conditions [2].

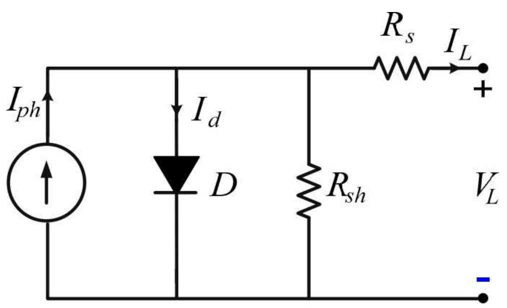


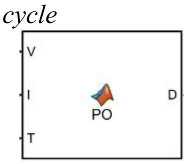
Figure 1. PV Equivalent Circuit

The photovoltaic equivalent circuit shown in Figure 1 is a simplified model representing the electrical behavior of PV cells. Maximum Power Point Tracking (MPPT) stands as a pivotal method for enhancing the efficiency of photovoltaic systems by capturing the utmost accessible power output. However, partial shading presents challenges for traditional MPPT methods like perturb and observe [3]. This study introduces a novel comparative MPPT strategy to enhance energy conversion efficiency in PV systems under partial shading. It overcomes drawbacks of conventional methods, resulting in improved performance and robustness [4].

- Perturb and Observe (P&O) Algorithm

The P&O algorithm, a popular MPPT method, iteratively adjusts a PV system's operating point to monitor power changes. However, it struggles to accurately follow the maximum power point (MPP) under partial shading conditions, attributed to its dependency on local power fluctuations for accurate monitoring [5]. The section provides a brief overview of the P&O and discusses its limitations.

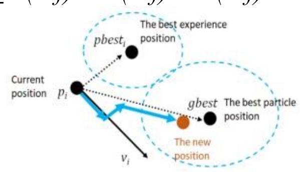
General Equations for P&O Algorithm;
If (Pnew > Pold): increase duty cycle
Else if (Pnew < Pold): decrease duty cycle
Else: maintain duty cycle



- Particle Swarm Optimization (PSO)

PSO represents a population-centric optimization method that derives inspiration from the collective motion of organisms, like when birds flock together or fish school. The two main characteristics of each particle are "Position and Velocity". Assessing an optimization's fitness function, which demonstrates how far a particle is from the ideal, or zero, optimal solution determines the particle's position [6]. Similarly to this, each particle's velocity determines how it moves in the solution space.

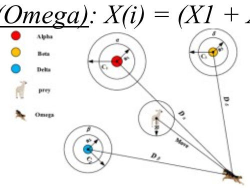
General Equations PSO Algorithm;
*<u>Velocity Update:</u> V(ref) = w * V(ref) + c1 * rand() * (Pbest(ref) - X(ref)) + c2 * rand() * (Gbest - X(ref)) + m * randn()*
<u>Position Update:</u> X(ref) = X(ref) + V(ref)



- Grey Wolf Optimization (GWO)

The GWO algorithm emulates grey wolves' hunting tactics for optimization problem-solving, particularly in MPPT for PV systems. This segment outlines GWO's core principles and its PV system MPPT application, while also examining its strengths and limitations. Its distinctive model structure sets it apart from alternative approaches. [7].

General Equations GWO Algorithm;
*<u>Leader (Alpha) Equation</u>: X1 = Xalpha - A1 * Dalpha*
*<u>Scout (Beta) Equation</u>: X2 = Xbeta - A2 * Dbeta*
*<u>Follower (Delta) Equation</u>: X3 = Xdelta - A3 * Ddelta*
<u>Position Update (Omega)</u>: X(i) = (X1 + X2 + X3) / 3

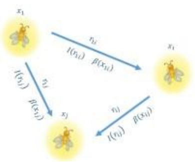


- Firefly Algorithm (FA)

The flashing behavior of fireflies served as the inspiration for the firefly algorithm, which is renowned for its ease of use and effectiveness in addressing optimization issues. This results in the ability of the entire population to naturally divide into subgroups, with each group having the ability to swarm around each mode or local optimum [8]. The most effective global solution can be found among all of these types. FA is superior to other algorithms in three key ways;

a. High ergodicity and diversity in solutions,
b. Automatically subdivision, and
c. The ability to deal with multimodality.

General Equations Firefly Algorithm;
*<u>Attractiveness</u>: beta(i,j) = beta0 * exp(-gamma * r(i,j)^2)*
*<u>Movement</u>: x(i,j) = x(i,j) + beta(i,j) * (x(j) - x(i)) + alpha * randn()*



To improve MPPT accuracy and robustness under partial shade situations, this comparative MPPT approach which combines the P&O algorithm with the PSO, GWO, and Firefly algorithms aims to make use of the benefits of each particular algorithm.

II. METHODOLOGY

The effectiveness of the suggested MPPT technique is evaluated using a comprehensive simulation-based experimental setup, employing a detailed model of a solar panel system. This model integrates pertinent mathematical formulations to portray the electrical behaviors of PV modules amidst changing shading scenarios, distinguishing itself by its distinct structural attributes from existing models. The configuration illustrated in Figure 2 represents a Simulink model that examines the organization of photovoltaic (PV) modules in an array setup of 3 rows and 3 columns, utilizing both series and parallel connections, as well as the influence of environmental factors and different partial shading scenarios [9]. This enables a thorough evaluation of the comparative MPPT approach's performance and its ability to optimize power generation under realistic conditions. The simulation-based experimental setup allows for a thorough analysis of the hybrid MPPT approach's performance and its ability to optimize power generation in the presence of shading conditions [10].

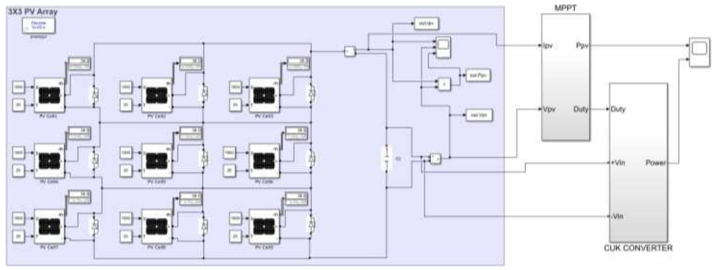
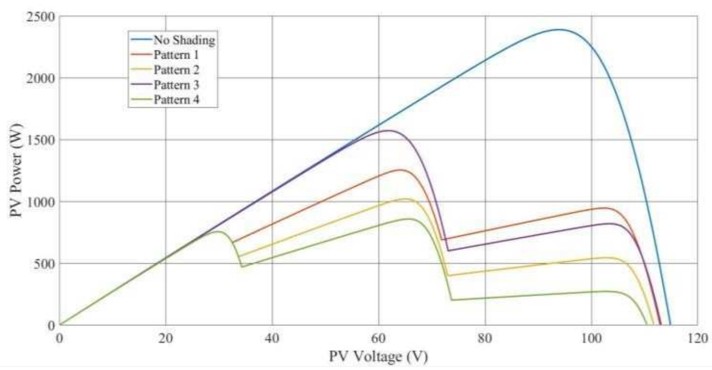


Figure 2. PV Simulink Model (Circuit Diagram)

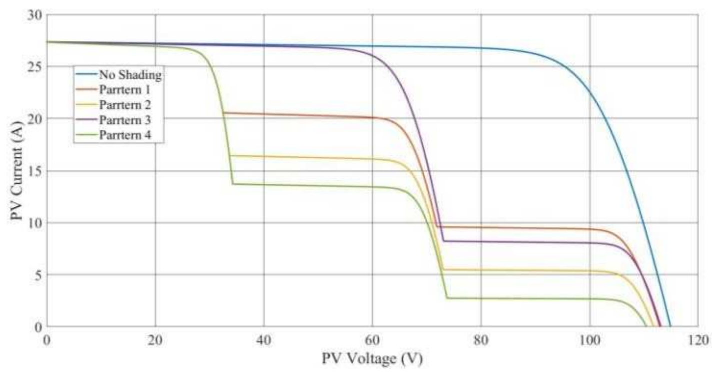
Table 1. Specification of PV (Photovoltaic) Panel

Model	User-defined
Peak Power	265.737W
Peak Voltage	31.3V
Peak Current	8.49A
Open Circuit Voltage	38.3V
Open Circuit Current	9.11A

Figure 3 (a - b) illustrates the P-V and I-V characteristics of a PV module under consistent 25°C temperature. The Maximum Power Point (MPP) of the PV module declines with decreasing solar irradiance, while it exhibits an upward trend as irradiance levels increase. These observations underscore the correlation between solar radiation and the PV module's MPP.



(a.)



(b.)

Figure 3. (a.) P-V, and **(b.)** I-V characteristics of PV modules at different shading patterns

The integration of P&O, PSO, GWO, and Firefly algorithms in the comparative MPPT technique enhances precision, convergence speed, and resilience in determining the MPP amidst partial shading [11]. Following iterative steps, the final voltage value for the estimated MPP is derived from the optimal fitness result achieved through PSO, GWO, and firefly algorithms. This voltage setting signifies the operational status for handling partial shading scenarios. The comparative MPPT procedure functions as outlined below;

The P&O algorithm is used to initially approximate the maximum power point (MPP) by making small voltage adjustments and measuring the resulting power changes [12]. It then uses these power variations to decide the appropriate direction for voltage adjustments, as illustrated in Figure 4.

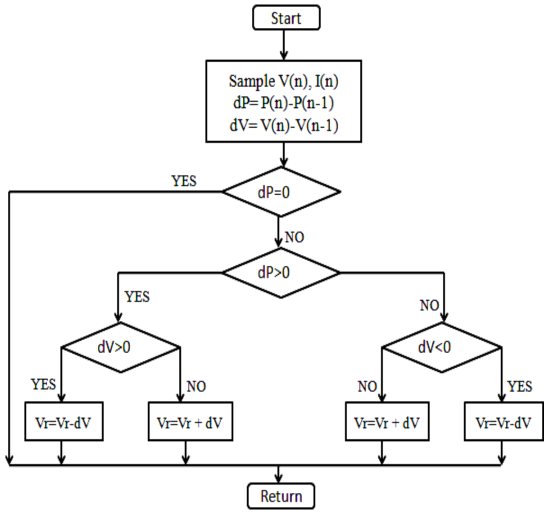


Figure 4. Flowchart of P&O

The utilization of the PSO algorithm enhances the accuracy of estimates generated by the P&O algorithm. Iteratively changing the voltage and power settings allows the swarm of particles to explore the search space [13]. The particles communicate their best solutions to update their velocities and positions, guiding the search toward the global MPP as illustrated in Figure 5.

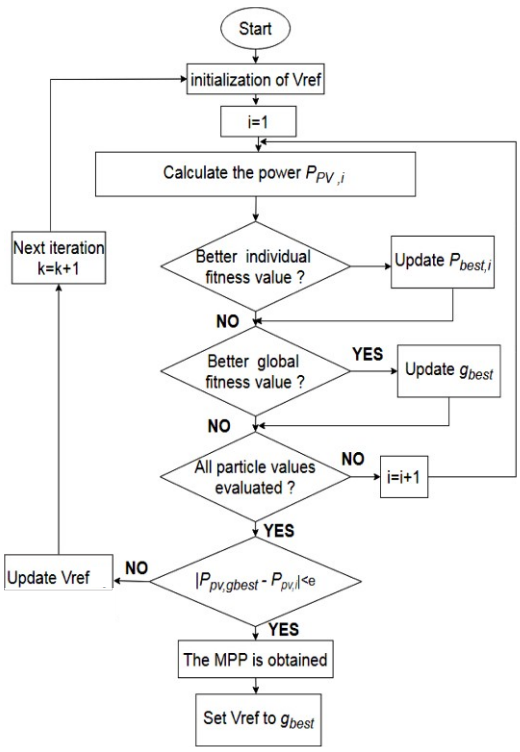


Figure 5. Flowchart of PSO

The GWO algorithm is used to increase MPPT accuracy even further. It explores the problem space using the hierarchical hunting strategies of grey wolves [14]. The process iteratively refines voltage optimization by updating alpha, beta, and delta wolf positions through fitness assessments as depicted in Figure 6.

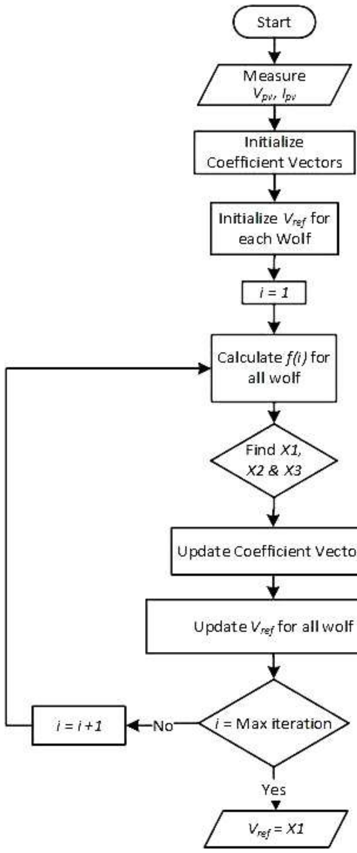


Figure 6. Flowchart of GWO

The Firefly algorithm is employed to enhance the robustness and stability of MPPT. It utilizes the attractiveness between fireflies to guide the search process as shown in Figure 7. The fireflies modulate their light emission levels in response to the effectiveness of their solutions, and this allure is employed to revise the voltage parameter [15].

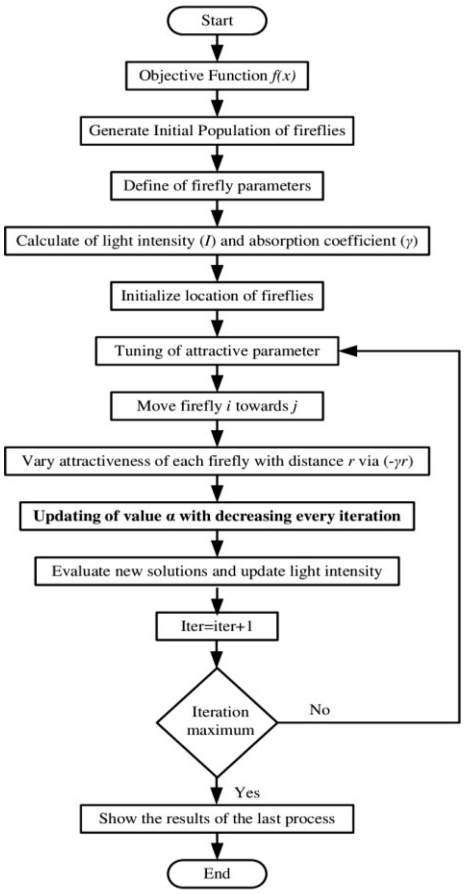


Figure 7. Flowchart of Firefly Algorithm

By conducting simulation experiments and analyzing the performance metrics, researchers can gain valuable insights into the strength and limitations of the proposed comparative MPPT approach. Different partial shading circumstances are applied to the PV system, and performance indicators like tracking efficiency, convergence speed, and stability are examined [16]. A comparative analysis is performed against individual MPPT techniques to demonstrate the superiority of the comparative approach.

III. Results and Discussions

This study evaluates three MPPT algorithms (PSO, GWO, and Firefly) combined with the perturb and observe technique to monitor the global maximum power point under uniform and irregular shading scenarios. MATLAB Simulink is used to develop the algorithmic and a DC-DC converter. The converter receives initial duty cycle values and adjusts them over based on the system's output power.

The non-uniform irradiation demonstrates how different shading patterns affect the GMPP as well as how each algorithm performs in monitoring the GMPP. The PV module's solar radiations under the first pattern of partial shading cases are $1000W/m^2$, $750W/m^2$, and $350W/m^2$. The power output of the GMPP stands at 1287.45 watts and is positioned at the second crest of the P-V curve, as depicted in Figure 8(a - d). While the PSO, GWO, and FA algorithms achieved convergence to the global maximum power point in 0.18 seconds, 0.137 seconds, and 0.125 seconds respectively, the P&O algorithm detected the GMPP in 0.132 seconds, yielding a power output of 1247 watts.

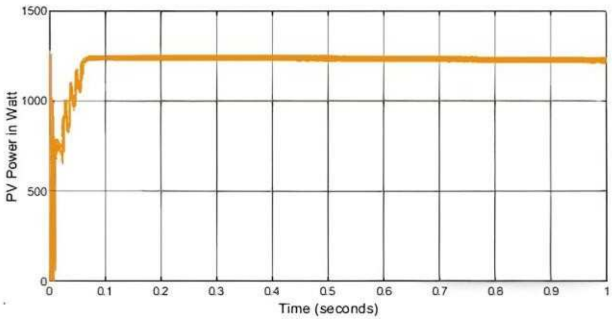


Figure 8(a). P&O

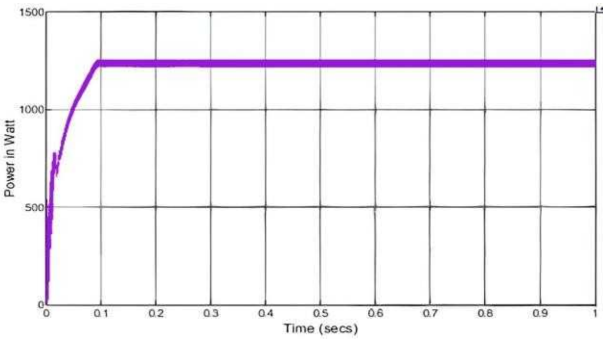


Figure 8(b). PSO

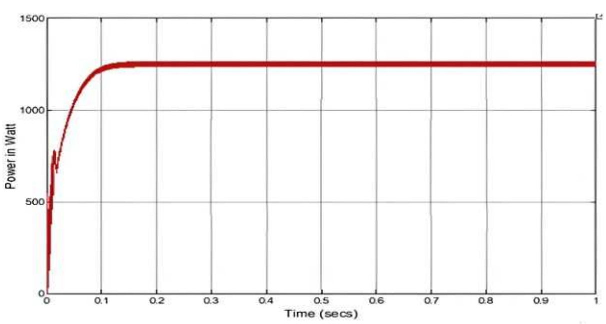


Figure 8(c). GWO

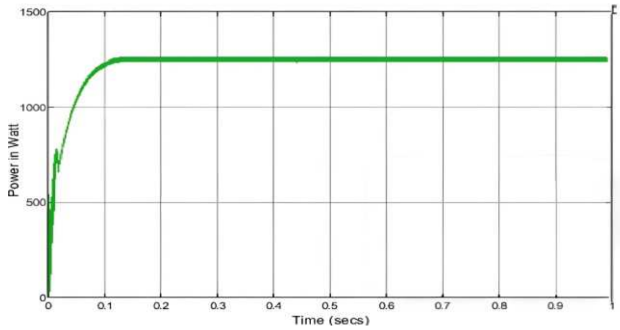


Figure 8(d). FA

Table 2. Outcomes of the first pattern

Algorithm	P_{pv} max. (W)	Duty Cycle	Track MPP	Tracking Time (sec)	Efficiency (%)
P&O	1247	0.342	Success	0.132	**96.86**
PSO	1253	0.36	Success	0.18	**97.32**
GWO	1261	0.359	Success	0.137	**97.94**
FA	1265	0.357	Success	0.125	**98.25**

The second configuration considers irradiance levels of $1000W/m^2$, $600W/m^2$, and $200W/m^2$. The P-V curve exhibits a second prominent peak at 1039.06W, representing the overall highest power output point for the second scenario. Figure 9(a - d) illustrates that PSO, GWO, and FA promptly detected the GMPP at 0.125s, 0.12s, and 0.113s, while P&O method did not achieve this.

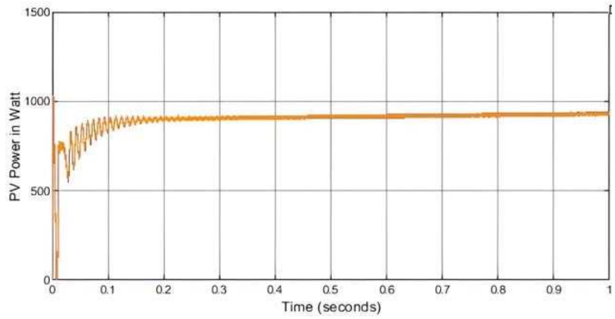


Figure 9(a). P&O

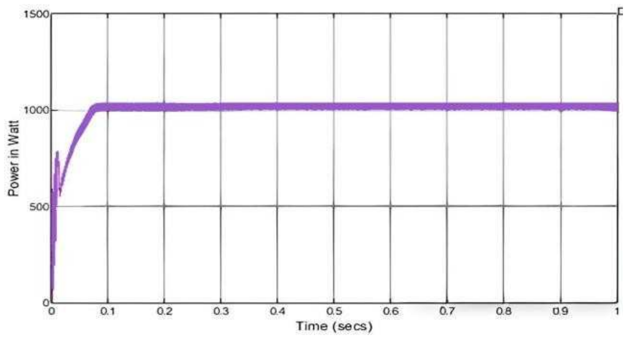


Figure 9(b). PSO

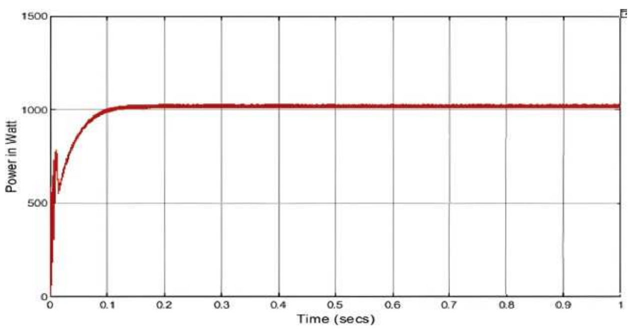


Figure 9(c). GWO

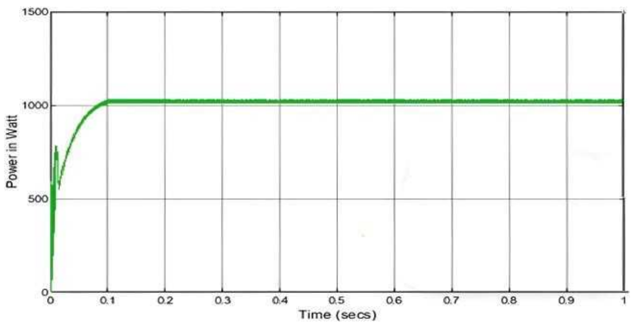


Figure 9(d). FA

Table 3. Outcomes of the second pattern

Algorithm	P_{pv} max. (W)	Duty Cycle	Track MPP	Tracking Time (sec)	Efficiency (%)
P&O	918	0.349	Fail	-	**88.35**
PSO	1021	0.34	Success	0.125	**98.26**
GWO	1023	0.33	Success	0.12	**98.45**
FA	1027	0.33	Success	0.113	**98.84**

The irradiations in the third pattern are (1000W/m², 1000W/m², and 300W/m²). Across the (P-V) graph, there are two peaks, the first peak is the GMPP and is equivalent to 1621.04W as shown in Figure 10(a - d). The attempt to implement the GMPP through the P&O approach did not result in its successful establishment, but the PSO, GWO, and FA algorithms did it in 0.12s, 0.114s, and 0.086s, respectively.

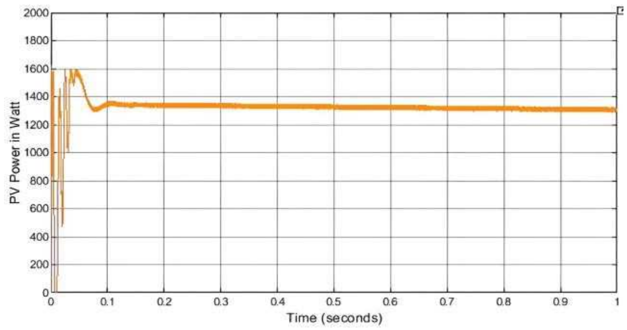


Figure 10(a). P&O

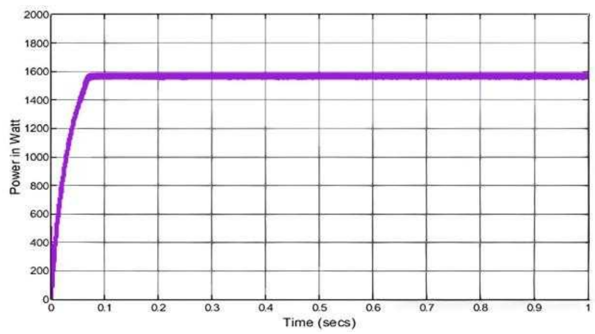


Figure 10(b). PSO

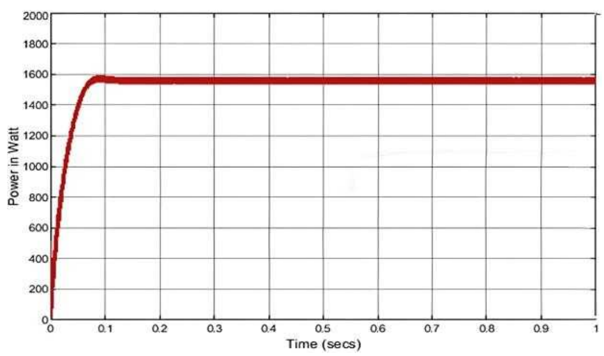


Figure 10(c). GWO

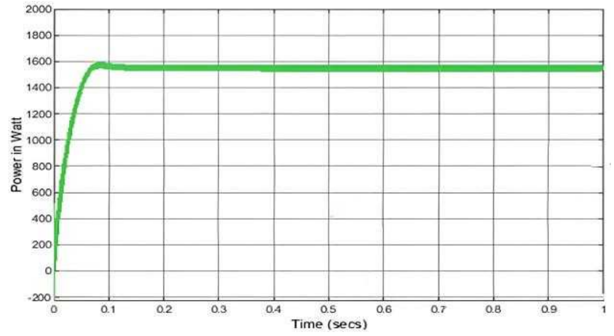


Figure 10(d). FA

Table 4. Outcomes of the third pattern

Algorithm	P_{pv} max. (W)	Duty Cycle	Track MPP	Tracking Time (sec)	Efficiency (%)
P&O	1342	0.346	Fail	-	**82.79**
PSO	1565	0.358	Success	0.12	**96.54**
GWO	1568	0.384	Success	0.114	**96.73**
FA	1584	0.388	Success	0.086	**97.71**

The solar irradiations were estimated to possess an identical value as (1000W/m², 500W/m², and 100W/m²) in the final pattern of investigation under partial shading. The point of highest value in the P-I curve's second peak, denoting the Gross Maximum Power Point (GMPP), registers at 875.64 watts. The global MPP may be identified and tracked successfully using FA, GWO, and PSO approaches shown in Figure 11(a - d). In comparison to PSO and GWO, it can be stated that the FA approach tracks GMPP more effectively and in a shorter amount of time.

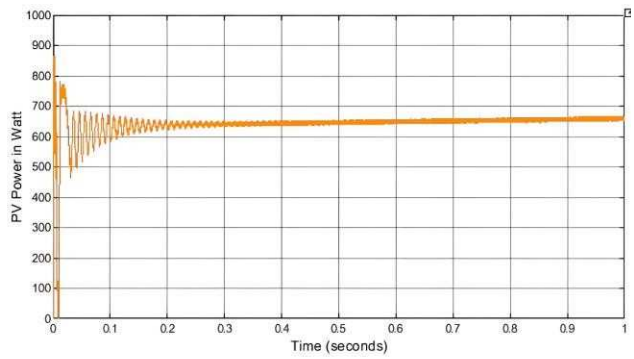


Figure 11(a). P&O

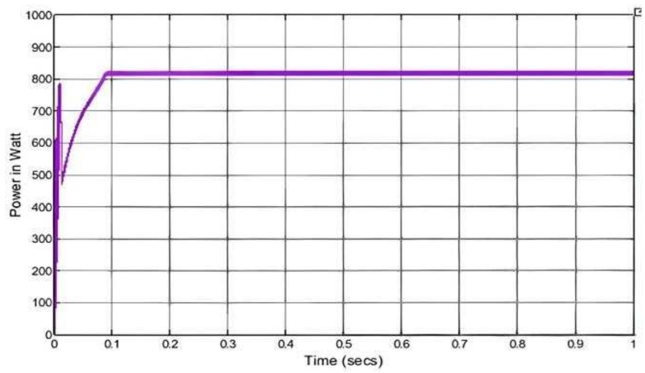


Figure 11(b). PSO

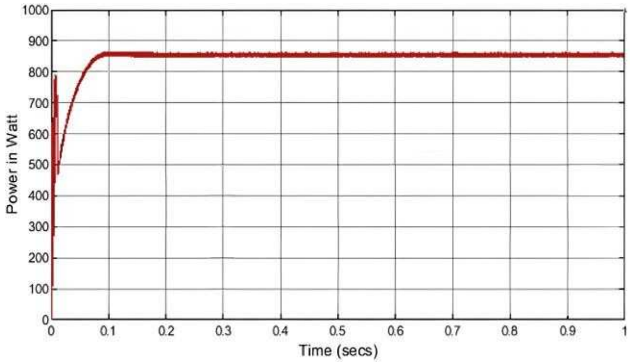


Figure 11(c). GWO

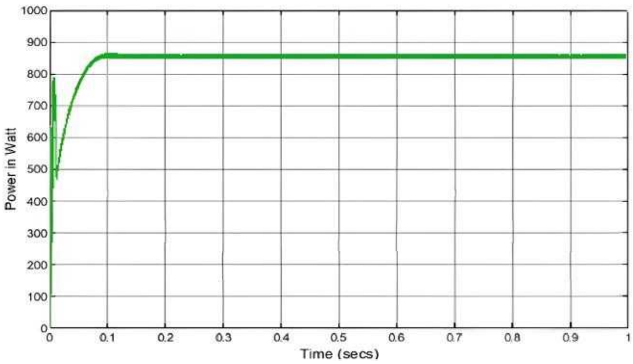


Figure 11(d). FA

Table 5. Outcomes of the fourth (final) pattern

Algorithm	P_{pv} max. (W)	Duty Cycle	Track MPP	Tracking Time (sec)	Efficiency (%)
P&O	648	0.346	Fail	-	**74.0**
PSO	837	0.321	Success	0.21	**95.58**
GWO	856	0.307	Success	0.14	**97.76**
FA	869	0.302	Success	0.098	**99.24**

In scenarios where shading is only partially present, the simulation findings presented in this work revealed that the GWO and FA algorithms achieved an average tracking efficiency of 97.72% and 98.51% respectively, accompanied by average tracking times of 0.127s and 0.105s respectively.

The outcomes of the study have been deliberated upon in the preceding theoretical analysis, the effect of shading on PV modules is significant and can impact their performance and efficiency. The average efficiencies achieved by **PSO, GWO, and FA are 96.925%, 97.72%, and 98.51%**, respectively.

IV. CONCLUSION

This study analyzed and contrasted the efficacy of four diverse maximum power point tracking techniques (P&O, PSO, GWO, and FA) for PV systems. The performance of these four algorithms was evaluated in scenarios where shading occurred on the solar panels. Enhancing heat storage diminishes potential savings by constraining the ability to shift solar heat gains from daytime to nighttime effectively. Having precise knowledge of the extent of shading is crucial in order to develop effective strategies for mitigating its adverse effects. Incorporating an aluminum reflector improves the effectiveness and efficiency of the photovoltaic (PV) system with only a marginal increase in expenses. The influence of partial shading can differ based on variables like the extent of shading and the frequency of its occurrence, and the arrangement of PV modules within the system. The influence of partial shading is contingent upon various factors such as shading level, occurrences, and PV module arrangement within the array.

The study assumed three radiation groups with constant fluctuations classified into four patterns. Compared to the other methods examined, the perturb and observe technique was less effective at identifying the global maximum power point, showing longer convergence times and higher oscillation levels. The remaining three algorithms, including the firefly algorithm, successfully located and tracked the GMPP. FA stood out for its speed, efficiency, and ability to converge, surpassing the performance of the other algorithms. Specifically, the FA method emerged as the most superior MPPT tracking approach across all tested scenarios, particularly when encountering increased shading on the panels.

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